

Week 2 Assignment 1:

CODE:

MySQL Local - Warning - not supported

Administration Schemas Query 1 SQL File 5* SQL File 6*

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Object Info Session

No object selected

```
1 CREATE DATABASE IF NOT EXISTS skytrack_db;
2
3 USE skytrack_db;
4 CREATE TABLE Flights (
5     flight_id INT AUTO_INCREMENT PRIMARY KEY,
6     flight_number VARCHAR(20) NOT NULL UNIQUE,
7     source VARCHAR(50) NOT NULL,
8     destination VARCHAR(50) NOT NULL,
9     departure_date DATE NOT NULL,
10    ticket_price DECIMAL(10,2) CHECK (ticket_price > 0)
11 );
12 CREATE TABLE Passengers (
13     passenger_id INT AUTO_INCREMENT PRIMARY KEY,
14     passenger_name VARCHAR(100) NOT NULL,
15     email VARCHAR(150) NOT NULL UNIQUE
16 );
17 CREATE TABLE Bookings (
18     booking_id INT AUTO_INCREMENT PRIMARY KEY,
19     passenger_id INT,
20     flight_id INT,
21     booking_date DATE,
22
23     FOREIGN KEY (passenger_id)
24         REFERENCES Passengers(passenger_id),
25
26     FOREIGN KEY (flight_id)
27         REFERENCES Flights(flight_id)
28 );
29 INSERT INTO Flights
30 (flight_number, source, destination, departure_date, ticket_price)
31 VALUES
32 ('ST101', 'Hyderabad', 'Delhi', '2026-10-01', 5500),
33 ('ST102', 'Hyderabad', 'Mumbai', '2026-10-02', 4200),
34 ('ST103', 'Hyderabad', 'Bangalore', '2026-10-03', 3500),
35 ('ST104', 'Delhi', 'Chennai', '2026-10-04', 6200),
36 ('ST105', 'Mumbai', 'Hyderabad', '2026-10-05', 4900),
37 ('ST106', 'Bangalore', 'Delhi', '2026-10-06', 5700),
38 ('ST107', 'Chennai', 'Mumbai', '2026-10-07', 5200),
39 ('ST108', 'Delhi', 'Bangalore', '2026-10-08', 5000),
40 ('ST109', 'Mumbai', 'Delhi', '2026-10-09', 4600),
41 ('ST110', 'Hyderabad', 'Chennai', '2026-10-10', 4100);
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

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Object Info Session

No object selected

```
41 INSERT INTO Passengers
42 (passenger_name, email)
43 VALUES
44 ('Rahul Sharma', 'rahul@gmail.com'),
45 ('Priya Reddy', 'priya@gmail.com'),
46 ('Arjun Kumar', 'arjun@gmail.com'),
47 ('Sneha Rao', 'sneha@gmail.com'),
48 ('Kiran Patel', 'kiran@gmail.com'),
49 ('Ananya Singh', 'ananya@gmail.com'),
50 ('Rohit Verma', 'rohit@gmail.com'),
51 ('Neha Gupta', 'neha@gmail.com'),
52 ('Vikram Das', 'vikram@gmail.com'),
53 ('Pooja Rani', 'pooja@gmail.com');
54
55 INSERT INTO Bookings
56 (passenger_id, flight_id, booking_date)
57 VALUES
58 (1,1,'2026-09-01'),
59 (2,2,'2026-09-02'),
60 (3,3,'2026-09-03'),
61 (4,4,'2026-09-04'),
62 (5,5,'2026-09-05'),
63 (6,6,'2026-09-06'),
64 (7,7,'2026-09-07'),
65 (8,8,'2026-09-08'),
66 (9,9,'2026-09-09'),
67 (10,10,'2026-09-10');
68 SELECT * FROM Flights;
69
70 SELECT * FROM Passengers;
71
72 SELECT * FROM Bookings;
73 SELECT
74     p.passenger_name,
75     f.flight_number,
76     f.source,
77     f.destination
78 FROM Bookings b
79 INNER JOIN Passengers p
80 ON b.passenger_id = p.passenger_id
81 INNER JOIN Flights f
82 ON b.flight_id = f.flight_id;
```

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

```

82      UN D.Flight_id = T.Flight_id;
83  * SELECT
84      destination,
85      COUNT(flight_id) AS Total_Flights
86  FROM Flights
87  GROUP BY destination
88  ORDER BY destination;
89  * CREATE TABLE Flight_History (
90      history_id INT AUTO_INCREMENT PRIMARY KEY,
91      flight_id INT,
92      action_description VARCHAR(255),
93      action_date DATE,
94      FOREIGN KEY (flight_id)
95          REFERENCES Flights(flight_id)
96  );
97  * START TRANSACTION;
98
99  * INSERT INTO Flights
100 (flight_number, source, destination, departure_date, ticket_price)
101 VALUES
102 ('ST111', 'Hyderabad', 'Kolkata', '2026-10-11', 5300);
103
104  * INSERT INTO Flight_History
105 (flight_id, action_description, action_date)
106 VALUES
107 (LAST_INSERT_ID(), 'New flight registered', CURDATE());
108
109  * COMMIT;
110  * SELECT * FROM Flights;
111
112  * SELECT * FROM Flight_History;
113  * CREATE INDEX idx_flight_number
114  ON Flights(flight_number);
115  * SELECT *
116  FROM Flights
117  WHERE flight_number = 'ST101';

```

100% 1:111

Query interrupted

OUTPUT:

Database + Tables created:

58	09:44:13	CREATE DATABASE IF NOT EXISTS skytrack_db	1 row(s) affected	0.0024 sec
59	09:44:13	USE skytrack_db	0 row(s) affected	0.00026 sec
60	09:44:27	CREATE TABLE Flights (flight_id INT AUTO_INCREMENT PRIMARY KEY, flight_number VARCHAR(20) NOT NULL UNIQUE, source...	0 row(s) affected	0.0080 sec
61	09:44:38	CREATE TABLE Passengers (passenger_id INT AUTO_INCREMENT PRIMARY KEY, passenger_name VARCHAR(100) NOT NULL, e...	0 row(s) affected	0.0058 sec
62	09:44:49	CREATE TABLE Bookings (booking_id INT AUTO_INCREMENT PRIMARY KEY, passenger_id INT, flight_id INT, booking_date DA...	0 row(s) affected	0.0085 sec

Running SELECT * FROM Flights:

	flight_...	flight_num...	source	destination	departure_da...	ticket_pri...
1		ST101	Hyderabad	Delhi	2026-10-01	5500.00
2		ST102	Hyderabad	Mumbai	2026-10-02	4800.00
3		ST103	Hyderabad	Bangalore	2026-10-03	3500.00
4		ST104	Delhi	Chennai	2026-10-04	6200.00
5		ST105	Mumbai	Hyderabad	2026-10-05	4900.00
6		ST106	Bangalore	Delhi	2026-10-06	5700.00
7		ST107	Chennai	Mumbai	2026-10-07	5200.00
8		ST108	Delhi	Bangalore	2026-10-08	5000.00
9		ST109	Mumbai	Delhi	2026-10-09	4600.00
10		ST110	Hyderabad	Chennai	2026-10-10	4100.00
11		ST111	Hyderabad	Kolkata	2026-10-11	5300.00
	NULL	NULL	NULL	NULL	NULL	NULL

Running SELECT * FROM Passengers:

	passenger_id	passenger_name	email	
	1	Rahul Sharma	rahul@gmail.com	
	2	Priya Reddy	priya@gmail.com	
	3	Arjun Kumar	arjun@gmail.com	
	4	Sneha Rao	sneha@gmail.com	
	5	Kiran Patel	kiran@gmail.com	
	6	Ananya Singh	ananya@gmail.com	
	7	Rohit Verma	rohit@gmail.com	
	8	Neha Gupta	neha@gmail.com	
	9	Vikram Das	vikram@gmail.com	
	10	Pooja Rani	pooja@gmail.com	
	NULL	NULL	NULL	

Running Bookings table

	booking_id	passenger_id	flight_...	booking_date	
	1	1	1	2026-09-01	
	2	2	2	2026-09-02	
	3	3	3	2026-09-03	
	4	4	4	2026-09-04	
	5	5	5	2026-09-05	
	6	6	6	2026-09-06	
	7	7	7	2026-09-07	
	8	8	8	2026-09-08	
	9	9	9	2026-09-09	
	10	10	10	2026-09-10	
	11	1	1	2026-09-01	
	12	2	2	2026-09-02	
	13	3	3	2026-09-03	
	14	4	4	2026-09-04	
	15	5	5	2026-09-05	
	16	6	6	2026-09-06	
	17	7	7	2026-09-07	
	18	8	8	2026-09-08	
	19	9	9	2026-09-09	
	20	10	10	2026-09-10	
	NULL	NULL	NULL	NULL	

Running Inner Join:

	booking_id	passenger_id	flight_...	booking_date	
	1	1	1	2026-09-01	
	2	2	2	2026-09-02	
	3	3	3	2026-09-03	
	4	4	4	2026-09-04	
	5	5	5	2026-09-05	
	6	6	6	2026-09-06	
	7	7	7	2026-09-07	
	8	8	8	2026-09-08	
	9	9	9	2026-09-09	
	10	10	10	2026-09-10	
	11	1	1	2026-09-01	
	12	2	2	2026-09-02	
	13	3	3	2026-09-03	
	14	4	4	2026-09-04	
	15	5	5	2026-09-05	
	16	6	6	2026-09-06	
	17	7	7	2026-09-07	
	18	8	8	2026-09-08	
	19	9	9	2026-09-09	
	20	10	10	2026-09-10	
	NULL	NULL	NULL	NULL	

Running GROUP BY + COUNT:

The screenshot shows the MySQL Workbench interface. The left sidebar contains navigation panels for Management, Instance, and Performance. The main editor displays a SQL query (lines 79-90) that joins Passengers and Flights tables, groups by destination, and counts flights. Below the query editor, the 'Result Grid' shows the output of the query, displaying destination names and their corresponding total flight counts.

```

79 INNER JOIN Passengers p
80   ON b.passenger_id = p.passenger_id
81 INNER JOIN Flights f
82   ON b.flight_id = f.flight_id;
83 SELECT
84   destination,
85   COUNT(flight_id) AS Total_Flights
86 FROM Flights
87 GROUP BY destination
88 ORDER BY destination;
89 CREATE TABLE Flight_History (
90   history_id INT AUTO_INCREMENT PRIMARY KEY,

```

destination	Total_Fligh...
Bangalore	2
Chennai	2
Delhi	3
Hyderabad	1
Kolkata	1
Mumbai	2

Running SELECT * FROM Flights;;

100% 1:111

Result Grid Filter Rows: Search Edit: Export/Import:

	flight_...	flight_num...	source	destination	departure_da...	ticket_pri...
1		ST101	Hyderabad	Delhi	2026-10-01	5500.00
2		ST102	Hyderabad	Mumbai	2026-10-02	4800.00
3		ST103	Hyderabad	Bangalore	2026-10-03	3500.00
4		ST104	Delhi	Chennai	2026-10-04	6200.00
5		ST105	Mumbai	Hyderabad	2026-10-05	4900.00
6		ST106	Bangalore	Delhi	2026-10-06	5700.00
7		ST107	Chennai	Mumbai	2026-10-07	5200.00
8		ST108	Delhi	Bangalore	2026-10-08	5000.00
9		ST109	Mumbai	Delhi	2026-10-09	4600.00
10		ST110	Hyderabad	Chennai	2026-10-10	4100.00
11		ST111	Hyderabad	Kolkata	2026-10-11	5300.00
	NULL	NULL	NULL	NULL	NULL	NULL

Running SELECT * FROM Flight_History::

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Limit to 1000 rows

```

102 ('ST111','Hyderabad','Kolkata','2026-10-11',5300);
103
104 • INSERT INTO Flight_History
105 (flight_id, action_description, action_date)
106 VALUES
107 (LAST_INSERT_ID(), 'New flight registered', CURDATE());
108
109 • COMMIT;
110 • SELECT * FROM Flights;
111
112 • SELECT * FROM Flight_History;
113 • CREATE INDEX idx_flight_number

```

100% 30:112

Result Grid Filter Rows: Search Edit: Export/Import:

history_id	flight_...	action_description	action_date
1	11	New flight registered	2026-09-19
	NULL	NULL	NULL

Running Search using flight number:

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Administration

Schemas

Query 1

SQL File 5*

SQL File 6*

MANAGEMENT

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Users and Privileges

Status and System Variable

Data Export

Data Import/Restore

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Dashboard

Performance Reports

Performance Schema Setup

Limit to 1000 rows

107

(LAST_INSERT_ID(), 'New flight registered', CURDATE());

108

109 • COMMIT;

110 • SELECT * FROM Flights;

111

112 • SELECT * FROM Flight_History;

113 • CREATE INDEX idx_flight_number

114 ON Flights(flight_number);

115 • SELECT *

116 FROM Flights

117 WHERE flight_number = 'ST101';

100%

31:117

Result Grid

Filter Rows: Search

Edit:

Export/Import:

flight_...	flight_numb...	source	destination	departure_da...	ticket_pri...
1	ST101	Hyderabad	Delhi	2026-10-01	5500.00